

SQL & DATABASES · DAY 4 OF 10

Designing a good database

Relationships, ER diagrams and normalisation

Thursday 8 October 2026

Today

- After today you can:
 - Spot the problems in a messy, spreadsheet-style table
 - Recognise one-to-many and many-to-many relationships
 - Draw a simple ER diagram
 - Normalise a table step by step (1NF, 2NF, 3NF)
 - Turn a design into CREATE TABLE statements

Today's schedule

Time	What we do
09:00-09:10	Warm-up
09:10-09:50	Theory (slides)
09:50-10:00	Break
10:00-10:55	Code-along, part 1
10:55-11:05	Break
11:05-12:00	Code-along, part 2
12:00-13:00	Lunch
13:00-14:30	Lab, part 1
14:30-14:45	Break
14:45-16:15	Lab, part 2
16:15-17:00	Walkthrough and wrap-up

Why design matters

- A badly designed database is like a messy room
- You can find things, but it takes forever, and things get lost
- Good design = each fact is stored in exactly ONE place
- Today we learn how to tidy up

The spreadsheet trap

- Anna is written three times, with two different emails. Which is right?
- Many products squeezed into one cell

order_no	customer	email	city	products
1	Anna Lindqvist	anna.lindqvist@...	Uppsala	Hoodie, Cap x2
3	Anna Lindqvist	anna.lindqvist@...	Uppsala	Socks x3
11	Anna Lindqvist	anna.l@...	Uppsala	Joggers x2

Three problems (anomalies)

Problem	What happens
Update problem	Anna moves: we must change 3 rows. Miss one = wrong data
Delete problem	Delete Maria's only order = Maria is gone too
Insert problem	Can't add a new customer until they buy something

Relationships between tables

Type	Means	Example
One-to-one (1:1)	One A has one B	One person, one passport
One-to-many (1:N)	One A has many B	One customer, many orders
Many-to-many (N:M)	Many A have many B	Orders and products

Many-to-many needs a middle table

- You can't put a list of products in one cell
- So we make a table in the middle: order_items
- Each row = one product in one order
- orders 1 - N order_items N - 1 products

order_id	product_id	quantity
1	1	1
1	3	2

ER diagram: a map of your database

- Boxes = tables. Lines = relationships. 1 and N = how many

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customers          orders          order_items          products
+-----+          +-----+          +-----+          +-----+
| customer_id | 1---N | order_id    | 1---N | order_id    | N---1 | product_id |
| first_name  |      | customer_id |      | product_id  |      | name       |
| ...         |      | order_date  |      | quantity    |      | price      |
+-----+          +-----+          +-----+          +-----+

```

Normalisation: tidying in 3 steps

- 1NF, 2NF and 3NF are three tidy-up steps
- Each step removes one kind of mess

Every column must depend on the key, the whole key, and nothing but the key.

1NF: one value per cell

- No lists in a cell
- Every row has a primary key

Bad	Good
products = 'Hoodie, Cap x2'	One row per product in order_items
phone = '070-111, 073-222'	A separate phones table

2NF: depend on the **WHOLE** key

- Only matters when the key is made of two columns
- In order_items the key is (order_id, product_id)
- product_name depends only on product_id, not on the order
- So product_name belongs in products, not in order_items

3NF: nothing but the key

- A column must not depend on another normal column
- If orders had customer_city, the city depends on the customer, not the order
- So city belongs in customers
- Test: 'If X changes, do I have to change many rows?'

From words to tables: a recipe

1. Read the description. Underline the nouns (things)
2. Each important thing becomes a table
3. Its details become columns. Pick a primary key
4. Find the relationships: 1:N or N:M?
5. N:M? Make a middle table
6. Draw the ER diagram, then write CREATE TABLE

Remember

- Store every fact in exactly one place
- 1:N uses a foreign key. N:M needs a middle table
- Draw an ER diagram before you build
- 1NF: one value per cell. 2NF: the whole key. 3NF: nothing but the key

Your turn (1/2)

1. Look at this table: student | phone_numbers | course1 | course2 | course3. List every problem you can find.
2. In order_sheet, which normal form does the products column break? How would you fix it?
3. A table has: order_id | customer_id | customer_email | order_date. Which column is in the wrong place? Why?
4. Music school (in pairs): 'Students take lessons from teachers. A lesson has a date, time, room and instrument. One teacher can teach many instruments.' Underline the things.
5. Find the relationships in the music school. Which are 1:N and which are N:M?

Your turn (2/2)

6. Draw the ER diagram for the music school.
7. Write the CREATE TABLE statements for the music school in a new file music.db.

Coming up tomorrow

**Our data is now spread over many tables.
Tomorrow we learn JOIN to put it back
together!**